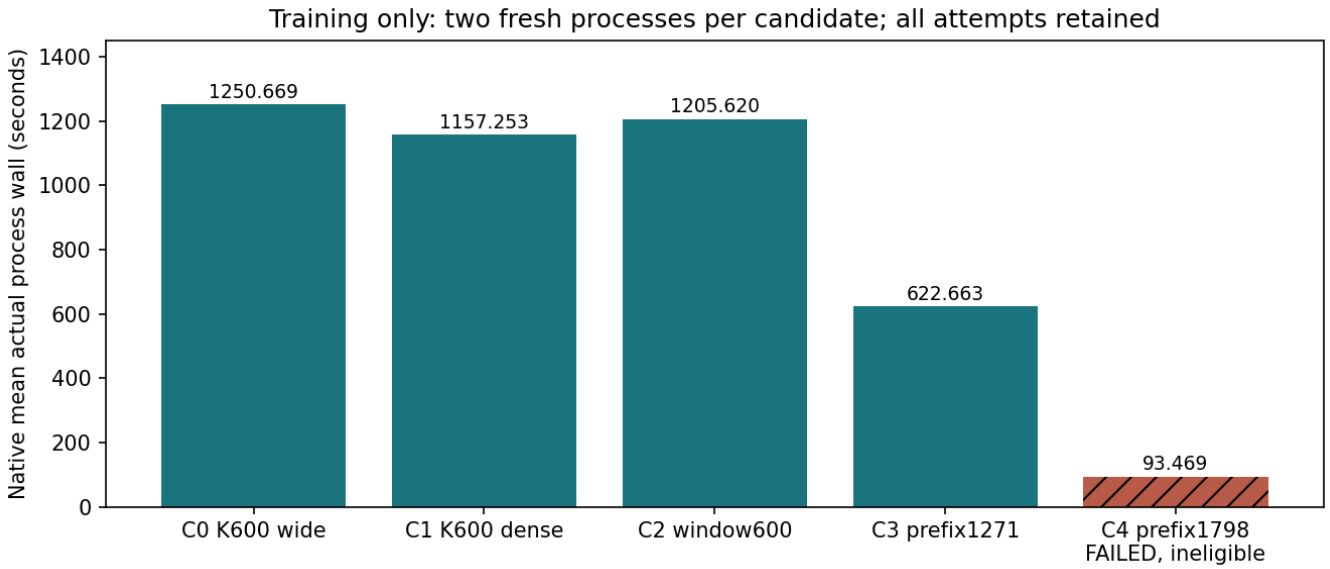
Degree-83 IC basis/setup: training stage

Run started 2026-10-15 15:16 UTC. Published worker/controller/protocol freeze: e07da4efdc092d6d7d1d7b660b29225a665d0775. Selection committed at 19664f02fdcb5f3e5674f5a26e989a117be2f0bc before fresh holdouts were generated.

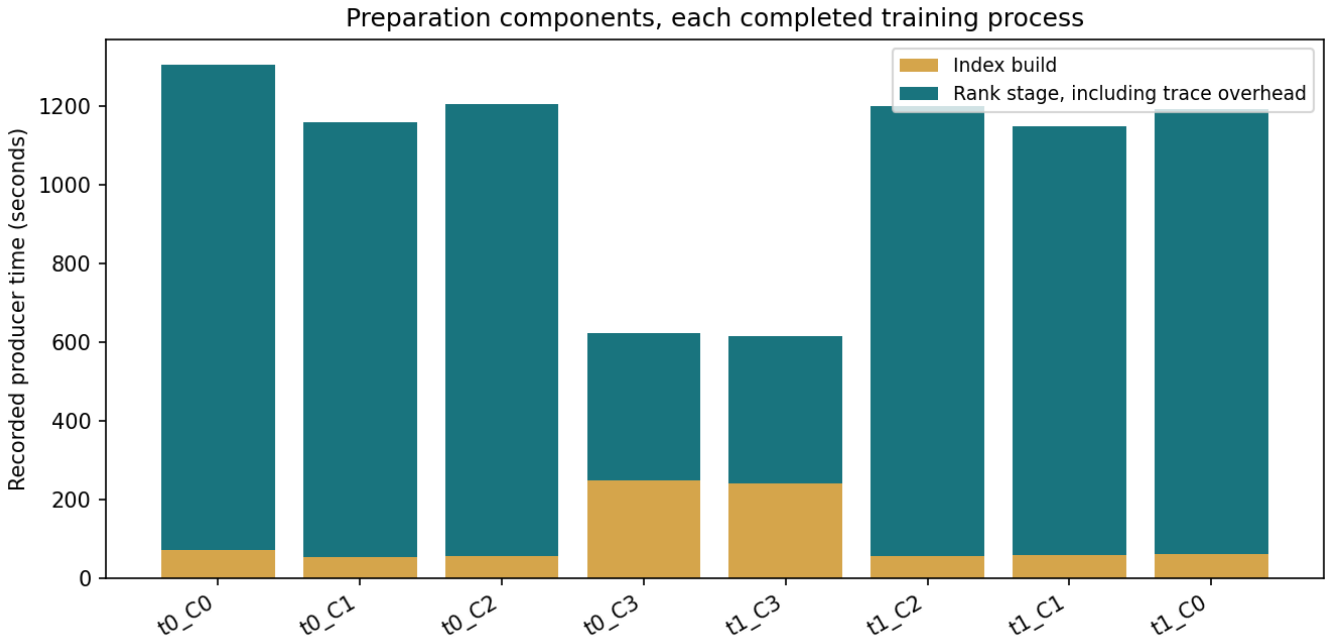
Observed: all 18 declared processes attempted; 16 completed and two C4 processes failed allocation. Eight completed IC transcripts passed independent C++ rank/log and scalar verification. Three rho A/A work pairs matched. C3 is selected by the predeclared training rule. No fresh-holdout end-to-end winner is established.



Means come directly from the native selector. C4 bars show failed-attempt cost and are ineligible; they do not indicate a fast solution. No uncertainty interval was declared for two-point training selection.

Exact legacy curve: a=b=1, GF(2^83) modulus z^83+z^7+z^4+z^2+1, subgroup r=8569786107849059; this is a 53-bit prime subgroup. Same CPU3, one OS thread, 12 GiB RLIMIT_AS and 2100-second complete-process cap for both methods.

What preparation costs



The native profile separates anchor construction, relation search excluding witness checks, witness checks, modular row construction, elimination and backsolve. Producer setup/rank clocks include instrumentation. They are diagnostics inside the actual complete-process wall, not replacements for it.

Run	Probes	Search (s)	Elimination (s)	Backsolve (s)
train_0_C0	1437855278	1234.456803	0.135536	0.001014
train_0_C1	1437855278	1105.871532	0.125664	0.000974
train_0_C2	1472146321	1150.648301	0.131778	0.000917
train_0_C3	507472060	371.262890	1.067147	0.004008
train_1_C3	507465286	373.928031	1.052403	0.003947
train_1_C2	1477593822	1145.210733	0.125019	0.001061
train_1_C1	1340076576	1090.793102	0.128799	0.001075
train_1_C0	1340076576	1130.508041	0.127625	0.001222

Every completed training run obtained one useful rank-gaining relation per attempt: 600 for C0/C1/C2 and 1271 for C3. These are producer counters whose exported modular rows/rank gains and four-point witnesses were independently replayed; search probes and durations were not independently rerun.

Complete process costs and failures

Process	Status	Wall (s)	CPU (s)	Peak RSS (bytes)
aa_0_0	PASS	4.630801	4.625734	62119936
aa_0_1	PASS	4.764793	4.686541	62115840
aa_1_0	PASS	2.018198	2.011767	19398656
aa_1_1	PASS	2.085859	2.081003	19464192
train_0_C0	PASS	1306.481963	1306.395022	5834244096
train_0_C1	PASS	1161.669895	1161.502747	2613014528
train_0_C2	PASS	1208.596824	1208.495951	2563850240
train_0_C3	PASS	626.157598	626.103073	10832527360
train_0_C4	FAILED	93.587788	93.559610	6588936192
train_0_rho	PASS	1.307987	1.305073	19456000
train_1_C4	FAILED	93.349774	93.330215	6588801024
train_1_C3	PASS	619.167648	619.114956	10832445440
train_1_C2	PASS	1202.643457	1202.541160	2563788800
train_1_C1	PASS	1152.836831	1152.742143	2613014528
train_1_C0	PASS	1194.857025	1194.758710	5834059776
train_1_rho	PASS	3.908712	3.901139	36507648
aa_2_0	PASS	4.435090	4.429189	62103552
aa_2_1	PASS	4.296349	4.290989	62119936

Both C4 stderr records say: memory allocation of 12884901888 bytes failed. Native process receipts retain the allocation failure, return/signal state and raw evidence. The memory-limit-hit indicator remains null; peak RSS is the wait4 whole-process measurement, not requested allocation size.

External checker wall/CPU/RSS is recorded per successful IC run in records/training-audit/audit_meter.json and excluded from solver costs. Both rho training singleton costs and all six A/A process costs remain visible above; no historic IC time is mixed with new rho time.

Scope, evidence and next stage

C0/C1 use the identical retained K600 mathematical base, isolating wide versus dense layout. C2 uses the next 600 public-scan orbits, verified disjoint from K600. Its entire 1200-orbit construction then discard/reindex is charged. C3/C4 enlarge the public prefix to 1271/1798. Dense packing is a representation change, not a new mathematical base.

An initial draft first-600 candidate equalled K600 and was corrected before timing. Two training points excluded 1911 old/base signed orbits. After the committed selection freeze, eight fresh points were generated outside 1913 orbits. Cold holdouts and three independently seeded eight-target preparation cohorts remain pending.

Public input validation, solve, final generic scalar replay and answer serialization/write/flush are included in matching aligned query clocks; full process wall/CPU/RSS remains primary. Original retained K600 construction and OS-cache flushing are outside scope. Endpoint VmRSS/HWM is not exclusive phase peak; trace timing does not calibrate all instrumentation overhead.

No 1685-target crossover is executed. No calibrated S, L2/independent-host admission, global optimum, asymptotic or deployed-security conclusion is claimed. Training selection is not holdout confirmation. All failed/preflight records remain preserved.

Protocol: ../../PROTOCOL.md

Selection: ../../selection/selection_freeze.json

Native profiles: native_profiles.json

Native profiler source: ../../postprocessing/profile_training.cpp

Audit receipts: ../../records/training-audit/audit_meter.json

Raw seals: ../../selection/TRAINING_RAW_SHA256SUMS

Study PR: https://github.com/aburan28/crypto/pull/1899. Required CI and human review remain merge gates. Report source plots recorded native JSON values; Python computes no selection, statistics or experimental result.